

Slot C

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Computer Vision

Course Code: ITA0502

1) Computer Vision is a branch of Artificial Intelligence (AI) that allows Computer to ~~analyze~~ analyze, process, and understand the images and videos.

a) * To capture images and videos

b) * To understand the pixel of image.

b) * Hospitals :- like x-rays.

* Phones :- Camera.

2) a) Feature extraction is the process of finding important parts of an image.

b) * Identify Object correctly.

* improves accuracy.

It has many features that helps Computer understand images accurately.

3) It process in three levels.

* Low level * Mid level * High level

a) Low-level - It improves the image quality by removing noise and adjusting brightness.

b) Mid level

Detects edges and shapes

produces image features

High-level

Recognize and understand objects.

produces decisions or information.

4) A digital image is made up of many tiny pictures elements called pixels. pixels together form the complete table.

a) A pixel is the smallest unit of digital image.

b) High resolution = More pixels = Better image.

Lower resolution = Fewer pixels = Blurry image.

5) Image acquisition is the process of capturing a real-world and converting it into digital values.

a) An image sensor captures the light and converts it into digital signals.

b) Digital camera, scanner.

6) Sampling:- Sampling and quantization converts an analog image into a digital image.

a) Sampling divides a image into small small pixels.

b) Quantization gives brightness or colour values to every pixel, to make the image digital image.

7) The image formation model explains how a camera captures a real-world scene and converts it into a image or video.

a) perspective projection converts a 3D objects into a 2D images.

b) Illumination provides light so the camera captures the clear image. Better lighting gives better image quality.

The image processing improves an image, while computer vision understands and analyzes the image.

a) image processing is the process of enhancing or modifying an image.

b) * image processing improves images; computer vision understands images.

* image processing gives an enhanced image. Computer vision gives information or decision.

The Computer Vision is used in many real-world applications.

a) Healthcare: It helps doctors to detect diseases from x-ray, MRI and CT scan images.

b) Automotive vehicles: It helps self-driving cars detect roads, traffic signs.

10)

The Digital image fundamentals provide the basic information needed for computer vision to analyze images accurately.

a) pixels are the smallest units of an image. Many pixels together form the complete image.

b) Sampling decides the number of pixels, High resolution provides more details.

